

Supplementary Information

S1. Scope and evidential status

This package reports an exploratory, staged computational study. D0, D1A, D1B, and D1C were separated by written designs, but no D2 confirmatory experiment was opened. All checkpoint, direction, block, fraction, and solver observations are nested within public pretrained-model artifacts whose complete randomization provenance is unavailable. The supplement therefore reports descriptive gates, not population-level hypothesis tests.

S2. Evidence classification

Statement	Class	Support	Ceiling
The pinned source contains 45,669 neurons, 1,513,231 edges, 64 types, 721 columns, and 734 fitted parameters	Fact	Source paper and inspected model	Fact about this artifact
Exact shams were zero and topology digests unchanged	Fact	Frozen result JSONs	Execution-path validation only
No G2 occurred among 45 eligible ladder cells	Fact	Complete fixed grid on checkpoints 008–009	Descriptive grid result; cells not independent
Eligible substitutions define an ordinary-response robustness region	Inference	Local pass plus ordinary/decoder G1 grid	Bounded to model, candidates, stimuli, checkpoints, margins
Mi1 timescale substitution altered causal gain	Inference	Cross-checkpoint/solver decoder-effect NRMSE; cosine ~1	Exploratory, fixed-interface, no matched-error confirmation
Local equivalence generally composes	Speculation	Not identified	Prohibited claim
Connectomes are biologically sufficient or insufficient	Speculation	Not tested	Prohibited claim
Consciousness or identity is preserved	Speculation	Not measured	Prohibited claim

S3. Source-grounded margins

Quantity	Source q10 standardized difference	One quarter	Frozen margin
Local block velocity	30.805%	7.701%	1.000%
Central block response	19.488%	4.872%	1.000%
Frozen decoder output	72.354%	18.089%	1.000%

The source ensemble contextualized but did not loosen the authored 1% ceiling. Checkpoint differences reflect model-artifact variation and are not biological repeatability.

S4. Local calibration

Family	Added capacity	Mi1	T4a	T5a	Decision
Timescale (alpha=.01)	no state; one shared coefficient	.908%	.716%	.956%	eligible all blocks

Family	Added capacity	Mi1	T4a	T5a	Decision
Two-state (beta=.25, tau=.1)	one scalar state/node; two shared coefficients	.985%	.613%	1.864%	T5a killed
Shunt (gamma=.01)	no state; one shared coefficient	.169%	.035%	.522%	eligible all blocks
Wrong shunt (gamma=1)	same form	failed	failed	failed	negative control separated

Values are aggregate derivative NRMSE. The actual gate also required both intensities and p99 ratios to pass.

S5. Global breadth accounting

Checkpoint	G0	G1	G2	Largest eligible full-block response	Largest eligible full decoder
008	2	21	0	.235%	.441%
009	1	24	0	.305%	.403%

G0 cells were retained as local failures and excluded from preservation/failure interpretation. Full ladder data are in `tables/table_global_ladder.csv`.

S6. Causal seam and solver veto

Checkpoint	dt	Ordinary NRMSE	Network-effect NRMSE	Mi1 block-effect NRMSE	Decoder-effect NRMSE	Effect cosine
008	.02	.0275%	1.484%	1.753%	2.331%	~1.000
008	.01	.0127%	.932%	.837%	2.754%	~1.000
009	.02	.0275%	1.326%	1.185%	1.897%	~1.000
009	.01	.0127%	1.020%	.809%	2.025%	~1.000

T4a timescale and two-state causal flags vanished at the smaller step. T5a timescale did not replicate across checkpoints. Only Mi1 × timescale remained cross-checkpoint and solver-robust, and only the decoder effect exceeded 1% in every cell.

S7. D1C matched-error stopping result

The matching target was timescale aggregate local NRMSE .909%. An admissible shunt required a complete local pass and an error ratio in [.8, 1.25].

gamma	Shunt NRMSE	Ratio	Complete pass	Outcome
.010	.152%	.167	yes	unmatched
.025	.380%	.418	yes	unmatched
.050	.761%	.836	no	intensity-1 NRMSE 1.027%
.075	1.141%	1.255	no	local failure
.100	1.521%	1.673	no	local failure

No post-outcome interpolation was permitted. The failed matching gate ended the generation, leaving checkpoints 013–014 untouched.

S8. Deviations and error log

1. An early checkpoint-001 timescale candidate ($\alpha = .02$) failed local transport after passing the preceding checkpoint and was killed; the margin was not relaxed.
2. An initial causal intervention hook had incorrect timing. The exact sham exposed the defect. The hook was corrected before accepting results, and the final exact errors were zero.
3. Several initial $dt = .02$ causal flags were killed by the $dt = .01$ solver veto.
4. D1C had an apparent near-match at $\gamma = .05$, but it failed the complete local gate. No interpolated rescue candidate was added.
5. No D2 confirmatory stage was opened.

S9. Statistical audit

- Independent unit intended: independently trained checkpoint.
- Actual limitation: public checkpoint metadata do not establish complete independent seed/randomization provenance.
- Nested observations: nodes, columns, frames, directions, intensities, replacement fractions, dynamics families, causal cells, and solver steps.
- Inferential p-values: none.
- Multiplicity correction: not applicable because no inferential family is claimed; the full frozen grid is reported.
- Missingness/exclusion: G0 cells were not dropped but classified as locally uninterpretable; crashes/nonfinite states would have counted as failures.
- Null interpretation: zero G2 in a fixed exploratory grid, not a universal null or frequency estimate.
- Hashes/tests: integrity and path-validation evidence only.

S10. Ethics and authorship

No living animals or human participants were used. The model is not a fly individual. Thomas Ryan is the sole human author and accountable operator. AI assistance is disclosed in the manuscript and does not constitute authorship or independent replication.